

ROSHAN KUMAR

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PROFESSIONAL SUMMARY

Software Engineer with production experience in enterprise Java/Spring Boot microservices for high-volume BFSI systems. Proven track record optimizing PL/SQL queries, architecting Business Rules Engines across 17+ financial products, and building event-driven platforms with Kafka and Kubernetes. Full-stack proficiency in React/Next.js with strong competitive programming foundation.

PROFESSIONAL EXPERIENCE

Systems Engineer (Java Developer)

Tata Consultancy Services

Oct 2024 – Present

Kolkata, India

- Engineered production-grade backend microservices for a large-scale insurance platform processing 100K+ daily policy transactions across multiple lines of business.
- Implemented and configured complex business logic within a centralized Business Rules Engine (BRE) across 17 financial products, automating dynamic premium calculations and underwriting validations that reduced policy change turnaround times by 60
- Refactored and optimized complex PL/SQL stored procedures handling multi-stage endorsement workflows, improving query execution time by 30–40% and eliminating bottlenecks in high-concurrency environments.
- Designed and implemented RESTful APIs for policy lifecycle operations (quote, bind, endorse, renew) with robust validation layers, reducing downstream errors by 25%.
- Collaborated with cross-functional teams (QA, DevOps, Business Analysts) in Agile sprints to translate ambiguous business requirements into scalable, maintainable Java/Spring Boot solutions.
- Tech Stack:** Java 11/17 · Spring Boot · Spring Data JPA · PostgreSQL · PL/SQL · REST APIs · Maven · Git · Postman

EDUCATION

Bachelor of Technology in Computer Science & Engineering

Vinoba Bhave University, Hazaribagh

2021 – 2024

TECHNICAL SKILLS

Languages: Java, JavaScript, TypeScript, C++, SQL (PostgreSQL, MySQL), PL/SQL

Backend & Frameworks: Spring Boot, Spring Data JPA, Node.js, Express.js, Hibernate

Frontend: React.js, Next.js, Redux, HTML5, CSS3, Tailwind CSS

Databases: PostgreSQL, MongoDB, MySQL

Messaging & Event Streaming: Apache Kafka, Socket.io

Cloud & Infrastructure: AWS (EKS, S3), Docker, Kubernetes, NGINX Ingress

DevOps & CI/CD: GitHub Actions, ArgoCD, GitOps, Maven

Tools: Git, Postman, Stripe API, WebRTC

PROJECTS

[Pizza Delivery Platform — Microservices & Event-Driven Architecture](#)

Production-grade, multi-tenant ordering platform with 5 independently deployed Node.js microservices communicating via **Apache Kafka**.

- **Auth Service:** PostgreSQL-backed RBAC (Admin/Manager/Customer) with JWT access + refresh token rotation; stateless validation across services eliminates single point of failure.

- **Event-Driven Caching:** Order Service maintains real-time product price cache by consuming Kafka events from Catalog Service — checkout validation happens locally without synchronous HTTP calls, reducing latency by 70%.
- **Payment Integration:** Stripe PaymentIntent flow with webhook signature verification; Socket.io broadcasts real-time order status updates to customers and managers.
- **Cloud-Native Deployment:** Backend on **AWS EKS (Kubernetes)** with NGINX Ingress; GitOps CI/CD via GitHub Actions (Docker → DockerHub) + **ArgoCD** for automated rollouts; Next.js/React frontends on Vercel.
- **Tech Stack:** Node.js · PostgreSQL · MongoDB · Kafka · Socket.io · Stripe · Kubernetes · Docker · ArgoCD · AWS S3 · Next.js · React

Source: github.com/roshan798/pizza-delivery-app

- **VoiceHub — Real-Time Group Voice Calling Application**

WebRTC-based group voice calling platform enabling multi-participant audio rooms with peer-to-peer connectivity.

- Built full-stack application using **React** (frontend), **Node.js/Express** (signaling server), and **MongoDB** for session persistence.
- Implemented **WebRTC** peer connections with **Socket.io** for real-time signaling (offer/answer SDP exchange, ICE candidate relay).
- Designed RESTful APIs for room management, user authentication, and call history tracking.
- **Tech Stack:** React · Node.js · Express · MongoDB · WebRTC · Socket.io

Source: github.com/roshan798/voicehub.me

ACHIEVEMENTS

- **LeetCode Contest Rating 1750+** — Participated in 56+ contests; ranked globally within top 1300 in Weekly Contest 376.
- **600+ Problems Solved** across LeetCode, GeeksforGeeks, and HackerRank with focus on algorithms, data structures, and system design.
- **HackerRank 6-Star** in Problem Solving demonstrating strong proficiency in competitive programming and algorithmic thinking.

CERTIFICATIONS

- [HackerRank Problem Solving](#)
- [Responsive Web Design](#)
- [Hacktoberfest Contributor](#)